



InsightAI dashboard

Project Proposal

Project Advisor:

Muhammad Zeeshan Nazar

Group Members:

Sohaib Ahmed	21L-5635
Muhammad Hamza Khan	21L-5654
Muhammad Hassan Khalid	21L-5692

National University Of Computer and Emerging Sciences
Department of Computer Science
Lahore, Pakistan

Abstract

In today's data driven world, the power to quickly analyze data is crucial. Existing analysis tools are complex, costly and lack flexibility making them a bad choice for users. This creates a gap, hindering effective data-driven decision-making. InsightAI dashboard fills this gap by offering a application that helps users in performing analytics with ease at no cost, designed to empower users by simplifying complex analysis tasks. Intuitive platform lets users gain insights from data without technical knowledge required. Application will support various file formats like CSV, SQLite databases and more. Famous libraries integrated to provide range of charts and analysis features. The AI-driven feature reduces time and effort required to analyze data allowing users to talk with data and gain information, helping them make informed decisions. Its power to streamline data visualizations, querying, and reporting makes it an everyday tool which improves user productivity and accessing full potential of their data.

1. Introduction

In today's world where data fuels progress, the power to quickly analyze and gain insights from data is of utmost importance to businesses and researchers. No one has the time to stare at vast amounts of data and try to generate useful insights from it. The options available right now in the market are paid, slow, complex for the new users and lack the flexibility option in them for the users. Our application InsightAI dashboard eliminates this problem by providing a powerful but still user-friendly and intuitive platform for data analysis that meets the needs of users with varying levels of expertise.

The application will feature a responsive and interactive interface for its users, making sure that the whole process could even be understood by a non-technical person. The backend of the application will be the area where the data processing and analysis will take place. It will feature popular libraries and frameworks for robust and quick performance.

Our application aims to automate and improve the way people carry out the data analytics tasks by using Artificial Intelligence (AI), Machine Learning (ML) and Natural Language Processing (NLP), eliminating the need for hiring expensive data analysts and engineers to extract useful insights from the data. By using our feature-rich platform, users will be able to talk with their data and get relevant information eliminating the need for expensive solutions. A key aspect of our application will be that users can select their desired columns for analysis, choose chart types and customize their visualizations without needing to have write lengthy codes. Our application aims to make the data analysis workflow smooth, from data import and cleaning to visualizing and reporting all within a single, integrated environment.

2. Goals and Objectives

Our aim is to make an optimized and feature-rich application that helps users to effortlessly generate useful insights from their data. The following are the key goals and objectives for the development of our application:

- Develop a intuitive and user-friendly web application for comprehensive data analysis.
- Develop a powerful backend system integrated with AI that can manage data analytics and processing tasks.
- Include interactive data analysis and visualization tools.
- Enable users to develop various types of diagrams such as Flowcharts, Sequence, Class, State, ER diagrams and many more directly within the application.
- Provide users the ability to generate and execute SQL queries within the application.
- Multiple options of various types of data preprocessing.
- Business analytics tools to allow users to gain valuable business numbers and key insights from data.
- Customizable interfaces and themes for the user interface.
- User guidance to explain the features and working of our application to users.
- Optimization for handling data effectively.
- Introduce a feature to generate data summaries and produce insightful reports answering key questions about the data.
- Feature of uploading documents and inquiring questions from the application so that users can get to know about their document without the need for going through it.
- Uploading various file formats including PDF, DOC, TXT, RTF and many more.
- Advanced statistical insights will be displayed such as correlation matrix and more for quick business analysis.
- Contextual memory will be implemented for a more personalized experience and accurate responses to the user.
- Develop a customizable dashboard within the application for users to create, modify, and export visualizations and charts directly.
- Allow users to customize visualizations and seamlessly export them into the integrated dashboard, with options to download in their preferred format.

By integration of the above features, we aim to make our application a powerful solution for transforming data into actionable insights.

3. Scope of the Project

Our application will consist of several components that will work harmoniously together to attain the project's desired goals. The front-end development will be focusing on creating such a user-friendly, responsive and intuitive interface that people of all technical and non-technical backgrounds can get a hold of the application and use it with ease. Our application will include a dashboard where users can easily organize, manage, edit and visualize their data, charts and graphs. Our application will allow support for uploading, processing, and storing various file formats including CSV, Excel, SQLite databases, and document types such as PDF, DOC, and TXT. We aim to implement natural language processing for the chat interface, context understanding and memory retention for personalized interactions with the application. The users will be able to add insights into the dashboard and manually edit their dashboards. The application will be powered by a suite of analysis tools such as SQL query generation, statistical analysis, and business analytics features. The backend development will be targeting the optimization of the data analysis to allow for handling of large data efficiently with a key concern on quick responses and processing. The application will be equipped with advanced algorithms and use popular libraries to generate the best insights and visualizations. There will be support for generation of various types of diagrams within the application itself such as Flowcharts, Sequence, Class, State, ER diagrams and many more. The application will also allow support for SQL querying, data pre-processing and cleaning. Furthermore, the application will allow users to do document analysis, customizable interface and seamless features will be added to export users work into different formats. Lastly, the application will also include data summary and data insights report generation from which users will be able to get key insight questions and their relevant answers.

Our application while designed for robust data analysis and superior performance still does have certain constraints such as that it has a token limit, it cannot process more than 15,000 tokens in a single interaction which may hinder the application in handling large datasets or very complex queries. Additionally, the application can only process datasets up to maximum 500 MB. It's also important to note that while the application is a very powerful tool that provides responses almost all the time users should be aware that on rare occasions it may produce errors in code or provide incorrect answers. Users should therefore approach the application with a close eye especially in situations where accuracy is paramount.

While ambitious in scope certain advanced features, for example real-time data connections, integration with external APIs, and collaborative editing will be considered for future development phases.

4. Initial Study and Work Done so Far

The initial phase of our application InsightAI dashboard has laid foundation stones for the project development. The project began with extensive data requirements gathering and analysis which was aided by the established methodologies and best practices by McKinney [1] and VanderPlas [2]. The research involved identifying user pain areas in the existing data analytic tools which helped us in identifying the features and requirements for our application.

To name a few, Microsoft Fabric [11], Power BI [12], and PowerApps [13] do not provide a intuitive user interface for new users leaving many unsure how to get started such as where to enter the data. Additionally, these services are paid and require separate subscriptions for Azure [16] cloud adding complexity and cost. Also, prior knowledge is required to run these platforms. Microsoft Copilot [15] and Google Colab [14] offer some data processing capabilities, but they perform poorly when it comes to data preprocessing tasks. ChatGPT [17] is better at responses but limited by a 250 MB dataset size which restricts its utility for larger data sets. Currently, there is no solution in the market that effectively addresses all these challenges.

Following this we identified different technological options available guided by the most recent developments in machine learning and data processing, as highlighted by Géron [3]. The chosen framework makes sure that our application is strong and accurate in handling complex data analysis tasks that the application is being designed to perform.

The architecture design phase was done with the help of principles of effective data visualization and dashboard design, as told by Few [4] and Tufte [5]. These principles helped in establishing the initial wireframes and user flow diagrams, making sure that the user interface is user-friendly, intuitive and responsive catering to both technical and non-technical users.

We also planned data visualization and SQL integration using helpful ideas from James et al. [7], Rajaraman and Ullman [8], and practical tips on data compression and mining from Salomon [9] and Witten et al. [10]. By doing so it made the data operations smoother and improved how we could interact with the data.

This in-depth foundation work makes sure that the application will be successfully developed. The next steps focus on expanding the functionality, improving the user interfaces and implementing core features based on feedback and testing.

References

- [1] W. McKinney, "Python for Data Analysis," 2nd ed. O'Reilly Media, 2017.
- [2] J. VanderPlas, "Python Data Science Handbook," O'Reilly Media, 2016.
- [3] A. Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow," 2nd ed. O'Reilly Media, 2019.
- [4] S. Few, "Information Dashboard Design: Displaying Data for At-a-Glance Monitoring," Analytics Press, 2013.
- [5] E. Tufte, "The Visual Display of Quantitative Information," 2nd ed. Graphics Press, 2001.
- [6] R. Chang, "R Graphics Cookbook," 2nd ed. O'Reilly Media, 2018.
- [7] G. James, D. Witten, T. Hastie, and R. Tibshirani, "An Introduction to Statistical Learning," Springer, 2013.
- [8] A. Rajaraman and J. D. Ullman, "Mining of Massive Datasets," Cambridge University Press, 2011.
- [9] D. Salomon, "Data Compression: The Complete Reference," 4th ed. Springer, 2007.
- [10] I. H. Witten, E. Frank, and M. A. Hall, "Data Mining: Practical Machine Learning Tools and Techniques," 3rd ed. Morgan Kaufmann, 2011.
- [11] "Microsoft Fabric" [Online]. Available: <https://www.microsoft.com/en-us/microsoft-fabric> [Accessed: 08-Aug-2024].
- [12] "Microsoft Power BI" [Online]. Available: <https://www.microsoft.com/en-us/power-platform/products/power-bi>. [Accessed: 08-Aug-2024].
- [13] "Microsoft Power Apps" [Online]. Available: <https://www.microsoft.com/en-us/power-platform/products/power-apps>. [Accessed: 08-Aug-2024].
- [14] "Google Colab" [Online]. Available: <https://colab.research.google.com/>. [Accessed: 08-Aug-2024].
- [15] "Microsoft Copilot" [Online]. Available: <https://copilot.microsoft.com/>. [Accessed: 08-Aug-2024].
- [16] "Microsoft Azure" [Online]. Available: <https://azure.microsoft.com/en-us>. [Accessed: 08-Aug-2024].

- [17] “ChatGPT - OpenAI” [Online]. Available: <https://openai.com/chatgpt/>. [Accessed: 08-Aug-2024].